$$e^{-\frac{\frac{1}{2}mv^2}{kT}\int\left(dv_x+dv_y+dv_z\right)^2}$$

$$= \quad dv_x^2 + dv_y^2 + dv_z^2$$

$$d\vec{v} = d\left(|v|^2 \sin\theta\right)$$

$$\left(e^{\lambda x^2}\right)' = 2\lambda x e^{\lambda x^2}$$

$$\int 2\lambda x e^{\lambda x^2} dx = e^{\lambda x^2}$$

$$\left(a x^n e^{\lambda x^m}\right)' = a n x^{n-1} e^{\lambda x^m} + a x^n e^{\lambda x^m} \cdot \lambda m x^{m-1}$$

$$= a n x^{n-1} e^{\lambda x^m} + a\lambda x^{n+m-1} e^{\lambda x^m}$$

$$\nabla^2 \varphi = 0$$

$$\nabla \times E = 0$$

$$\nabla \times B = -\mu_0 \varepsilon_0 \frac{\partial}{\partial t} E.$$

$$\nabla \times (\nabla \cdot E) = \nabla \times \left(\frac{1}{\varepsilon_0} \frac{\partial}{\partial t} B \right)$$

$$= \frac{1}{\varepsilon_0} \frac{\partial}{\partial t} \nabla \times B$$

$$= -\mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2} E$$

$$\psi'' = e^{\psi} - 1$$

$$\psi = \ln \phi$$

$$-\frac{1}{\phi^2} \phi'' = \phi - 1.$$